

IDLCV Project III: Image Segmentation

Group 24

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1. Introduction

This project has two main components. The first analyzes two CNN architectures: a baseline Encoder-Decoder and the U-Net. Both are trained and evaluated on the medical imaging benchmarks DRIVE (retinal vessels) and PH2 (skin lesions). This section covers dataset analysis, model implementation, performance metrics, and a loss function ablation study (BCE, Focal Loss, Weighted BCE).

The second part addresses model training under weak supervision, a pragmatic constraint given the high cost of dense annotations. We trained a U-Net using only sparse, simulated user "clicks." This section analyzes the click generation algorithm and presents an ablation study on click quantity and balance.

2. Part 1 - Image segmentation

In the first part, we are asked to implement architecture for two different datasets. For each of them, first we develop an analysis to check how the datasets are built to understand the different tasks. Then, the implementation of the two requested models is done for both datasets, after which we train them and evaluate with different metrics. Finally, an ablation study is performed to determine the loss function with which the models perform best.

2.1. Dataset analysis

2.1.1. DRIVE dataset

The DRIVE (Digital Retinal Images for Vessel Extraction) dataset consists of 40 retinal fundus images organized into training and test sets. Each image in the training set contains the original retinal fundus photograph (e.g., `21_training.tif`) and the corresponding manual vessel segmentation mask (e.g., `21_manuall.gif`) located in the `1st_manual/` folder. Additionally, field-of-view masks are provided in the `mask/` folder, which indicate the circular region of interest but were not used for vessel segmentation as they do not contain vessel information.

The dataset structure revealed a critical constraint: only 20 images (numbered 21-40) have ground truth vessel annotations available for training and evaluation, while the remaining 20 images (numbered 01-20) are provided without annotations as they were originally designed for competition evaluation with private ground truth.

Given the limited availability of annotated data, we deviated from the original DRIVE competition split and created our own train/validation/test division. The split was performed by randomly selecting from the 20 available annotated images, resulting in 70% for training, 15% for validation, and 15% for testing. This yielded 14, 3, and 3 images respectively for each split, ensuring we could properly evaluate model performance across all experimental phases.

This custom splitting approach was necessary because using the original test set (images 01-20) would have prevented any performance evaluation, as ground truth masks are not publicly available for these images.

2.1.2. PH2 dataset

This dataset is formed of 200 folders that each contains the original image of skin with a lesion (*IMDXXX_Dermoscopic_Image*), the segmentation of the skin lesion (*IMDXXX_lesion*) and a folder named *IMDXXX_roi* which for some images contains regions of interest. We have not used the latest as not all images contained images in this folder.

The split of the dataset was performed by randomly taking 70% of the images for training, 10% for validation and 20% for testing; resulting in 140, 20 and 40 images, respectively, for each split. For all the images, a resizing to 256x256 pixels is performed to maintain high resolution while not having high memory consumption.

2.2. Models and training

For both datasets, we developed two models:

- CNN Encoder-decoder CNN (EncDec): the first one consists of 5 downsampling layers with 64 features, one

bottle-neck CNN layer and another 5 upsampling layers. The downsampling layers consist of one CNN layer, a *ReLU* activation function and *Max Pool* to reduce size of the image by a half in each dimension. In the other hand, a upsampling layer consists of an upsample to duplicate the size in each dimension, a CNN layer and a *ReLU* activation function. The last upsampling layer does not include the activation function to get the desired output.

- Unet: This model is similar to the previous one but it includes the skip connections; these skip connections consist of concatenations of the output of the encoder layers with the input of the corresponding decoder layers. It also increments the number of features while we reduce the dimensionality in the downsampling and reduces the features while increasing the dimensionality in the upsampling.

For both models, the training is performed with the *Adam* optimizer with a learning rate of 0.0005 and including early-stopping to avoid overfitting with 10 epochs of patience. In addition, the selected loss function is a combination of dice loss and BCE, giving a weight of 0.5 to each of them. For the evaluation, we tested the model with several metrics, including: Dice overlap, Intersection over Union, Accuracy, Sensitivity, and Specificity.

2.2.1. DRIVE dataset

In Figure 1 we see that the U-Net model converges much faster than the simple encoder-decoder model and achieves significantly lower training and validation loss. The skip connections in the U-Net architecture enable more effective learning of vessel segmentation patterns, as evidenced by the smoother convergence curves and superior final performance.

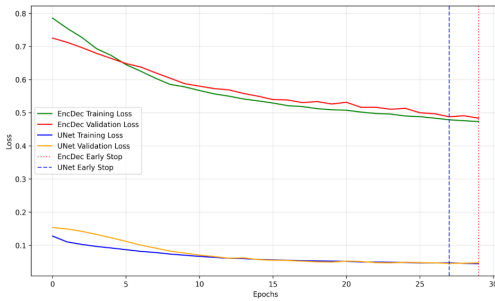


Figure 1. Training and validation loss for both models.

In Table 1 we can observe that the U-Net model obtains better results in general than the EncDec model, with the U-Net achieving superior performance across all evaluation metrics, demonstrating the effectiveness of skip connections for retinal vessel segmentation.

Metric	Encdec	U-Net
Dice Coefficient	65.52%	72.16%
IoU	48.90%	56.46%
Accuracy	94.57%	95.27%
Sensitivity	58.47%	70.68%
Specificity	98.08%	97.62%

Table 1. Metrics of the models for DRIVE dataset

2.2.2. PH2 dataset

We can clearly observe in the figure 2 that the Unet model converges much faster than the EncDec model and obtains a lower training and validation loss.

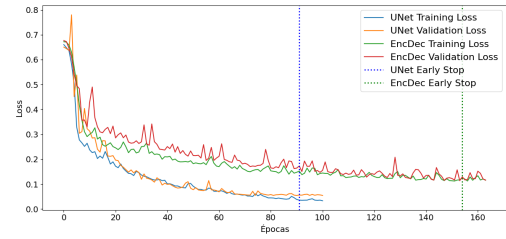


Figure 2. Training and validation loss for both models.

In the table 2 we can observe that the Unet model obtains better results in general than the EncDec model, although the values are not much bigger, obtaining both really good results.

Metric	EncDec	Unet
Dice overlap	94.34%	95.56%
IoU	89.32%	91.58%
Accuracy	96.72%	97.30%
Sensitivity	94.17%	96.45%
Specificity	97.82%	97.73%

Table 2. Metrics of the models for the PH2 dataset

2.3. Ablation study: loss functions

Finally, we performed an ablation study on the loss function for the Unet model, training with Binary Cross Entropy, Focal Loss and Weighted BCE. To calculate the weights, we obtained the frequency of the pixels of the background and the pixels of the object to segment and normalize so the frequency sum 2 (number of classes).

2.3.1. DRIVE Dataset

In Table 3 we can find the results of the metrics for the different loss functions. We quickly see that Focal loss obtains the best results and Weighted BCE performs slightly better than standard BCE, with Focal loss significantly outperforming both other approaches.

Metric	BCE	Focal loss	Weighted BCE
Dice overlap	57.61%	71.75%	60.62%
IoU	40.61%	56.04%	43.53%
Accuracy	93.48%	95.38%	93.67%
Sensitivity	49.39%	65.58%	53.64%
Specificity	97.81%	98.33%	97.53%

Table 3. Ablation study on DRIVE dataset.

2.3.2. PH2 dataset

In the table 4 we can find the results of the metrics for the different loss functions. We quickly see that Focal loss obtains the worst results and BCE and Weighted BCE obtain similar values, with BCE slightly better.

Metric	BCE	Focal loss	Weighted BCE
Dice overlap	97.04%	90.25%	96.06%
IoU	94.30%	82.40%	92.43%
Accuracy	97.87%	94.66%	97.71%
Sensitivity	96.59%	84.01%	95.36%
Specificity	98.66%	99.13%	98.48%

Table 4. Ablation study on PH2 dataset.

3. Part 2 - Weak annotations

3.1. Generation of clicks

We implemented an algorithm that simulates user interactions on segmentation masks to train models that learn from sparse annotations. **Positive clicks** are generated inside the mask regions. The algorithm computes the Euclidean distance transform of the mask to find pixels farther from the boundaries and samples a random number of click points (within a given range). Sampling is weighted so that central pixels are more likely to be chosen. To ensure full coverage, at least one click is placed in each connected component of the mask, and random jitter is added to make the clicks less regular. **Negative clicks** are generated outside the mask. The code samples points both near and far from the lesion. Near points are weighted inversely by distance to the mask (closer pixels more likely), while far points are chosen uniformly in the background. Random jitter is applied again.

3.2. Dataset and architecture

We trained the Unet model using images from the PH2 dataset along with the automatically generated clicks described above. For validation and testing, however, we used full segmentation masks. To support this setup, we adapted the loss functions so they could accept both clicks and complete masks as inputs.

3.3. Ablation study: number of clicks

We vary the number of positive and negative clicks per image according to different pairs. For each configuration, we report the mean Dice coefficient, mean Intersection over Union (IoU), and mean pixel accuracy on the test set. Table 5 summarises the results.

Pos	Neg	Mean Dice	Mean IoU	Accuracy
2	2	0.8418	0.7484	0.8894
4	4	0.8021	0.6824	0.8718
6	6	0.8417	0.7411	0.8919
10	10	0.8600	0.7649	0.9076
5	2	0.8282	0.7279	0.8794
8	3	0.8302	0.7267	0.8867
12	4	0.8583	0.7653	0.9044
4	12	0.8246	0.7163	0.8818
3	6	0.7978	0.6820	0.8684
4	8	0.8119	0.7003	0.8801
6	9	0.8338	0.7270	0.8907
9	1	0.8207	0.7085	0.8903
1	9	0.7358	0.6108	0.8386

Table 5. Ablation on the number of positive and negative clicks per image.

The ablation reveals three main trends:

- **Total number of clicks.** Increasing the total number of clicks generally improves performance. The best Dice scores are obtained for high-interaction settings such as (10, 10) and (12, 4), both around Dice $\approx$ 0.86.
- **Balance between positives and negatives.** For a similar total number of clicks, configurations with more foreground (positive) clicks tend to outperform those dominated by background clicks. For example, (12, 4) clearly outperforms the strongly background-heavy (4, 12), and (9, 1) is much better than (1, 9).
- **Effect of negative-heavy setups.** Strongly negative-biased configurations, such as (1, 9), lead to a noticeable drop in Dice and IoU, indicating that an excessive number of background clicks is less informative for guiding the segmentation compared to foreground clicks.

Overall, the study suggests that interactive performance benefits from a moderate to high number of clicks, and that allocating a larger fraction of clicks to the foreground region is preferable to heavily emphasising background clicks.

4. Conclusion

With this project we learned the U-Net’s architectural strength and offered insights into loss function trade-offs and the potential of weakly-supervised methods, which are crucial for reducing the annotation bottleneck in medical image analysis.